



VISUAL ANALYTICS: INFORMATION VISUALIZATION 101

AB MOSCA (THEY/THEM)



TODAY'S PLAN

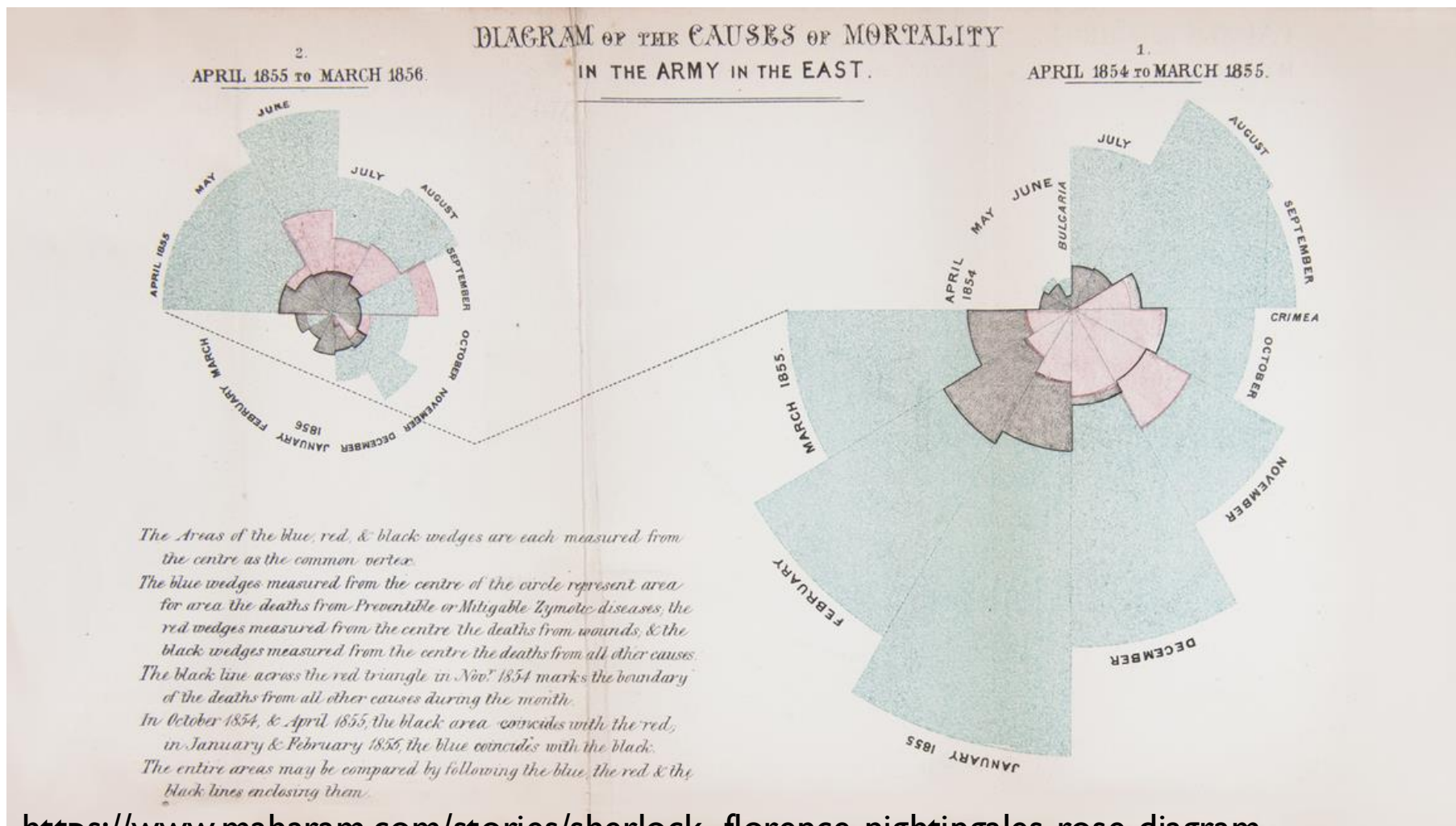
- What is information visualization
- Marks and channels
- Rules of thumb
- Reading visualizations

WHAT IS INFORMATION VISUALIZATION?



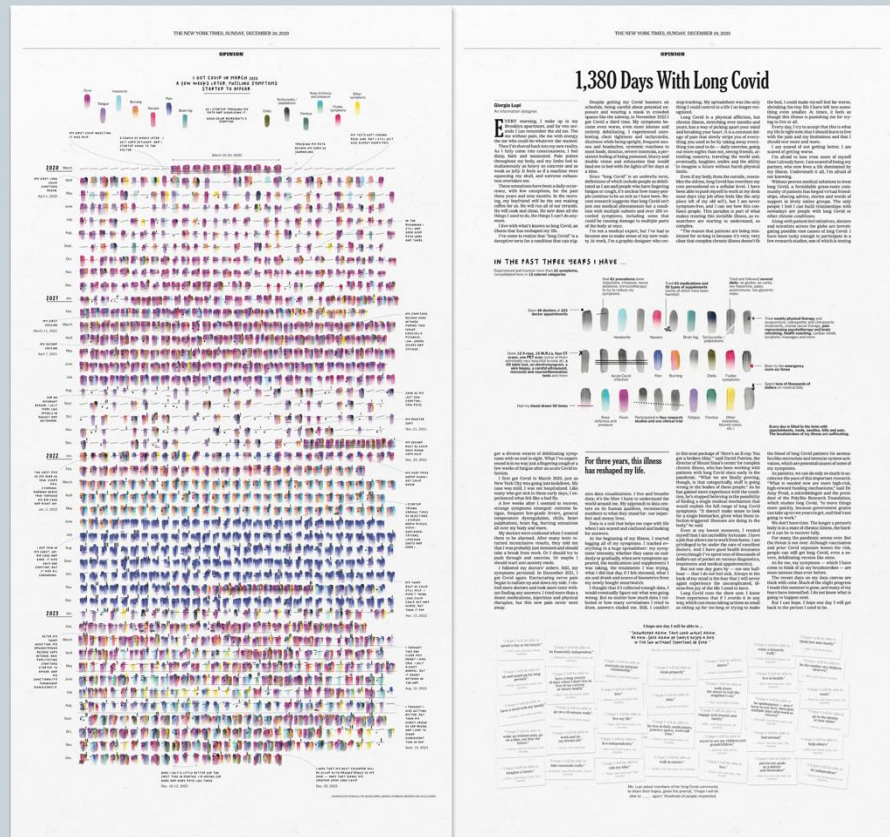
Something that reveals a trend?

WHAT IS INFORMATION VISUALIZATION?



Something that influences decision making?

WHAT IS INFORMATION VISUALIZATION?



Something that tells a story?

WHAT IS INFORMATION VISUALIZATION?

- **Visual representations** of data that **reinforce human cognition**

WHAT IS INFORMATION VISUALIZATION?

- Visual representations of data that reinforce human cognition

country	year	cases	population
Afghanistan	1999	7745	19987071
Afghanistan	2000	8666	20095360
Brazil	1999	37737	17206362
Brazil	2000	80488	17404898
China	1999	212258	127291272
China	2000	215766	128042583

variables

country	year	cases	population
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observations

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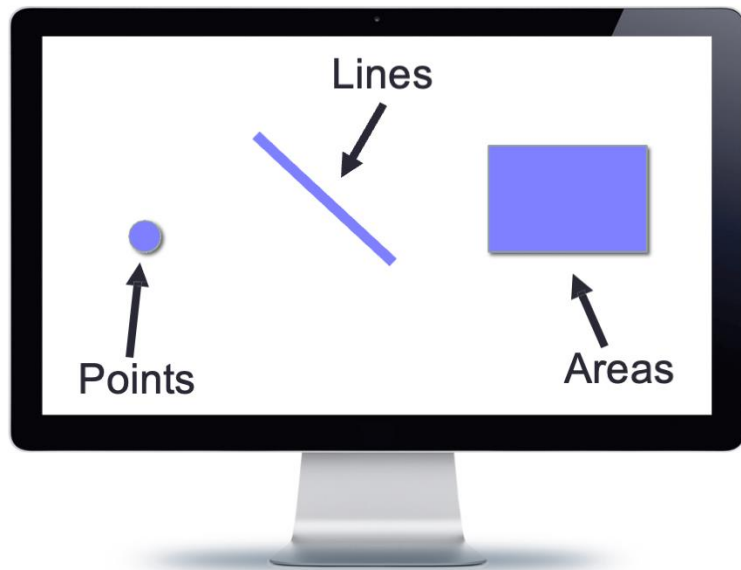
values



MARKS & CHANNELS

■ Visual representations of data that reinforce human cognition

■ Composed of:



- **Marks** – the “ink”
- **Visual Channels** – *how* the marks show up on the page

Spatial region



Color hue



Motion



Shape



Position on common scale



Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



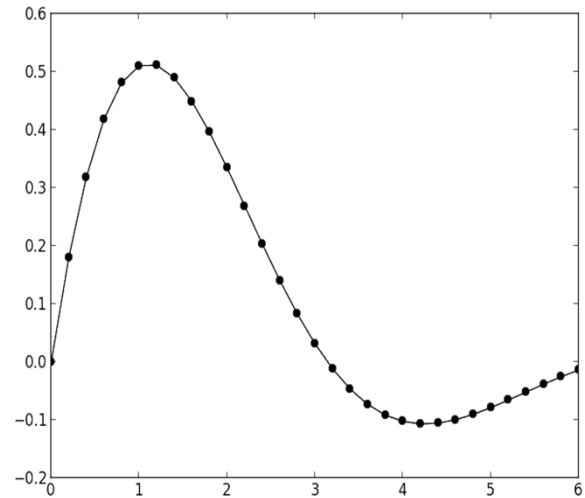
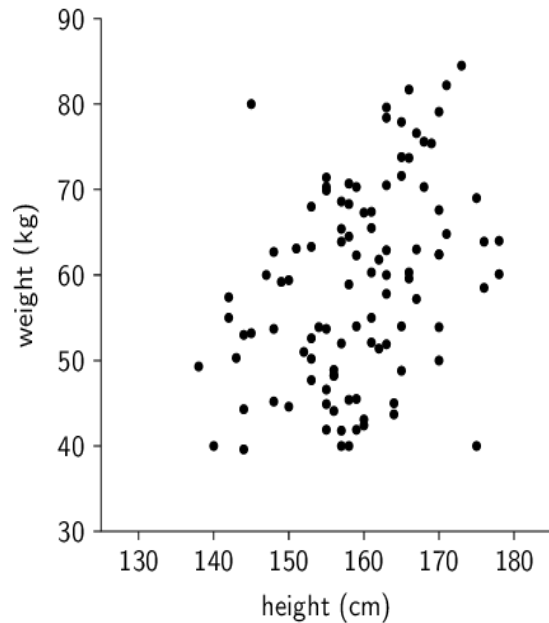
MARKS & CHANNELS

- **Visual representations** of data that **reinforce human cognition**
 - Typically, **one mark represents one observation** in the dataset
 - For each observation, the **values** of the variables describing that observation **are encoded via visual channels**

MARKS & CHANNELS

■ Visual Channels

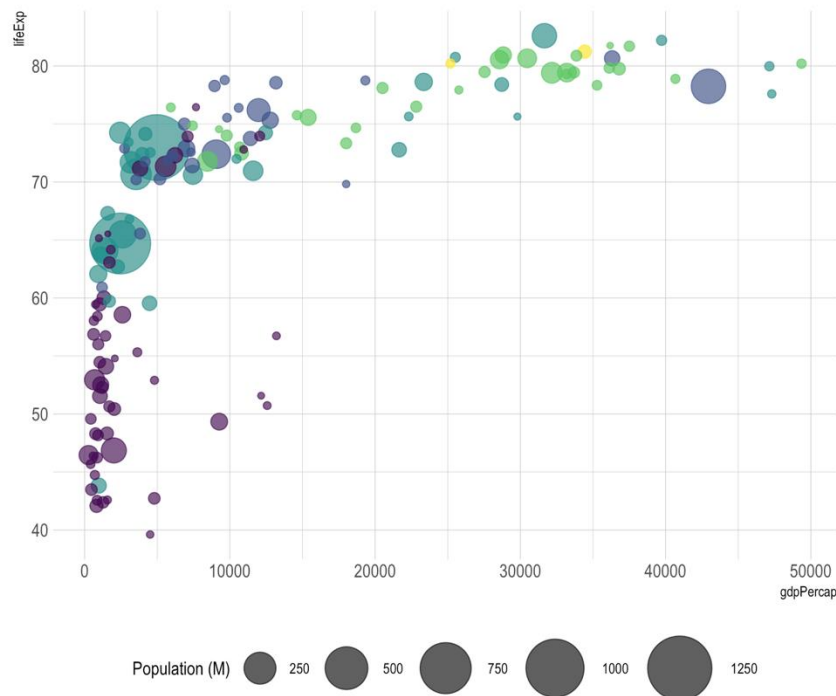
- Position / spatial region of a mark encodes value using *where* the mark is drawn



MARKS & CHANNELS

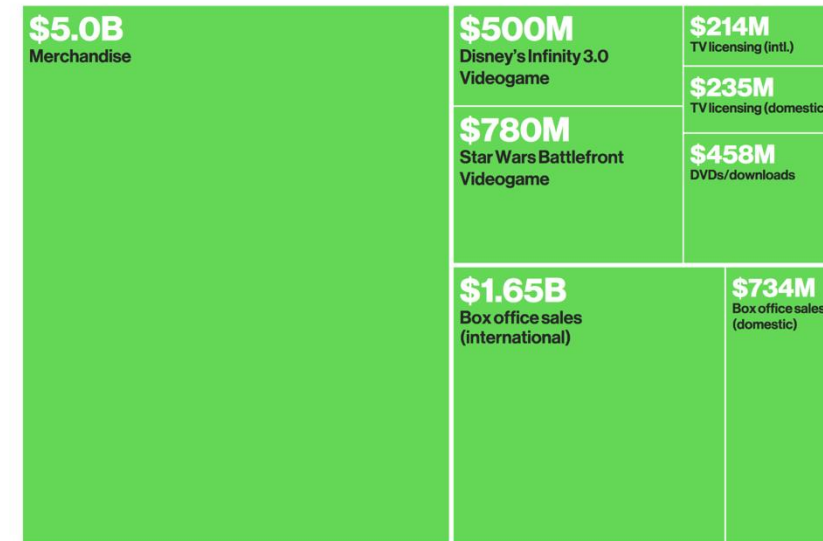
■ Visual Channels

- Size of a mark encodes value using *how big* the mark is drawn



<https://www.data-to-viz.com/graph/bubble.html>

"The Force Awakens" could produce **\$9.6 billion** in revenue from worldwide ticket sales, merchandise, and home entertainment in roughly the first year of release.

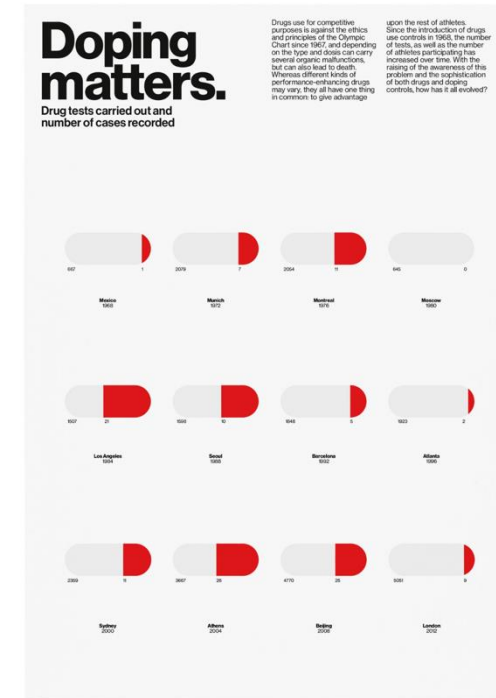
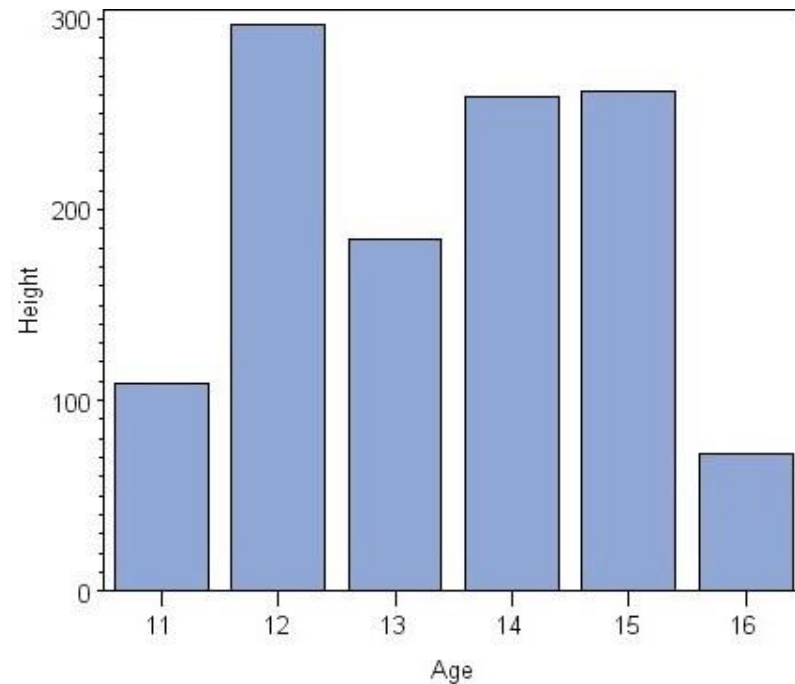


<https://www.bloomberg.com/graphics/2015-star-wars-force-awakens-revenue/>

MARKS & CHANNELS

■ Visual Channels

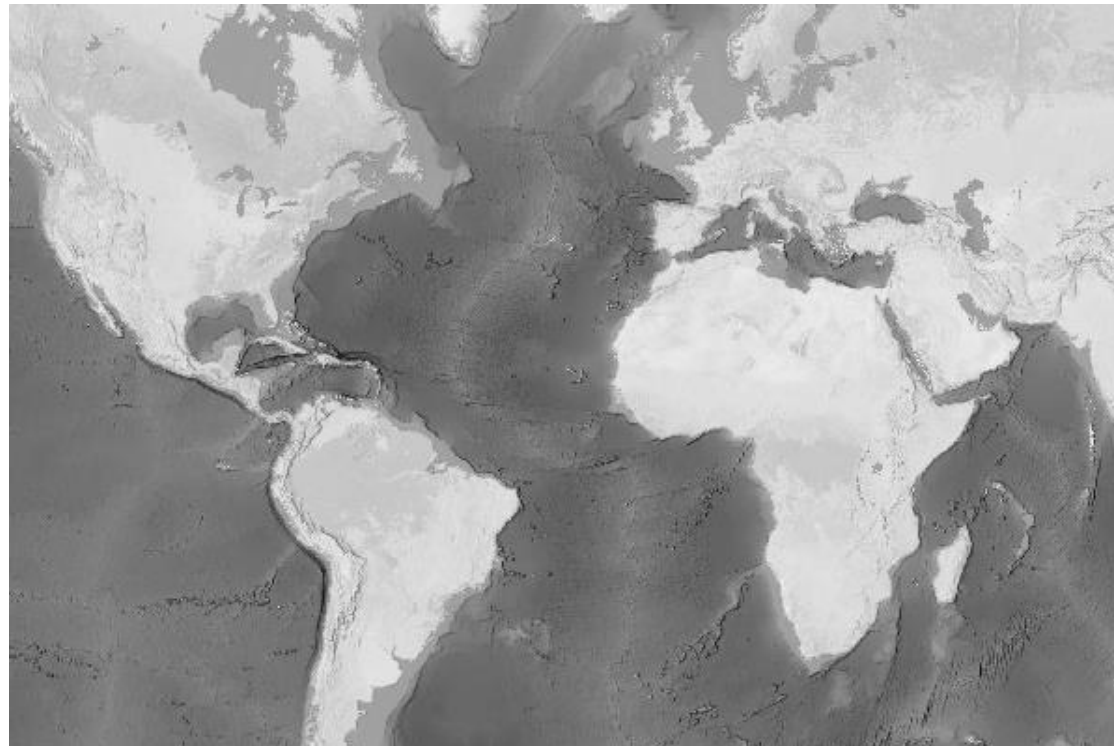
- Length of a mark encodes value using *how long* the mark is drawn



MARKS & CHANNELS

- **Visual Channels**

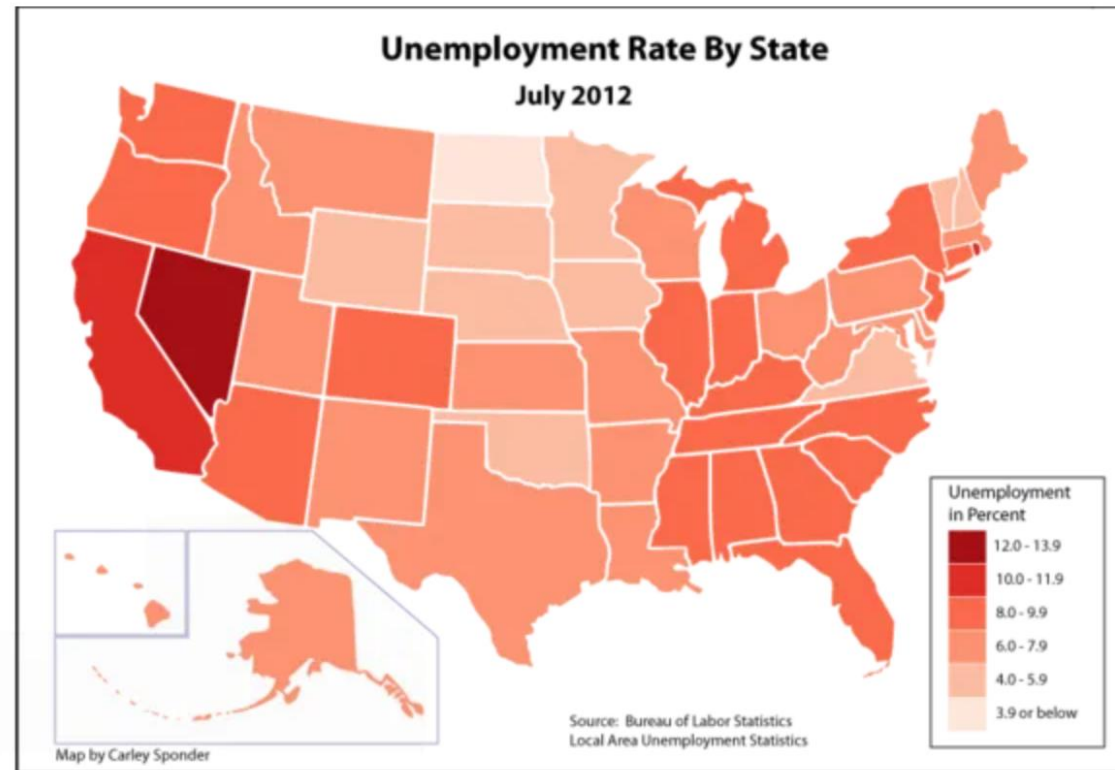
- **Luminance** of a mark encodes value using *how dark* the mark is drawn



MARKS & CHANNELS

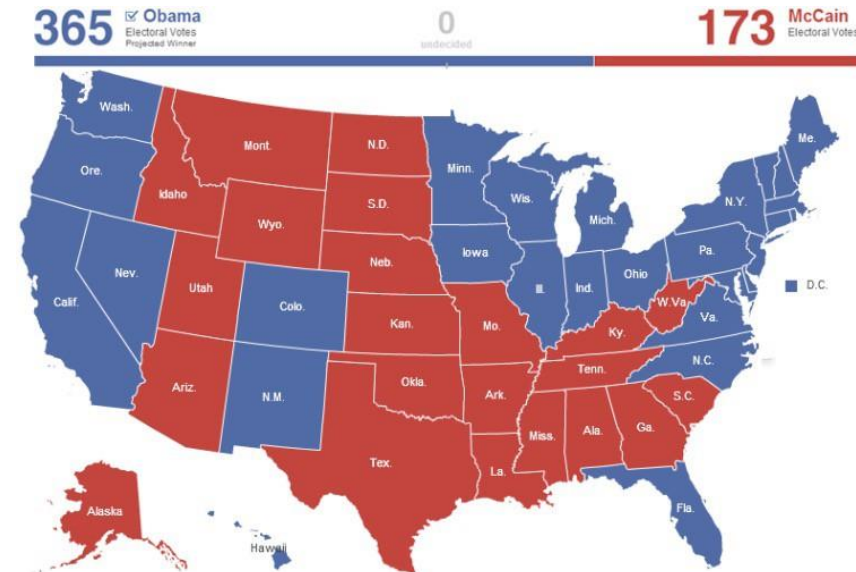
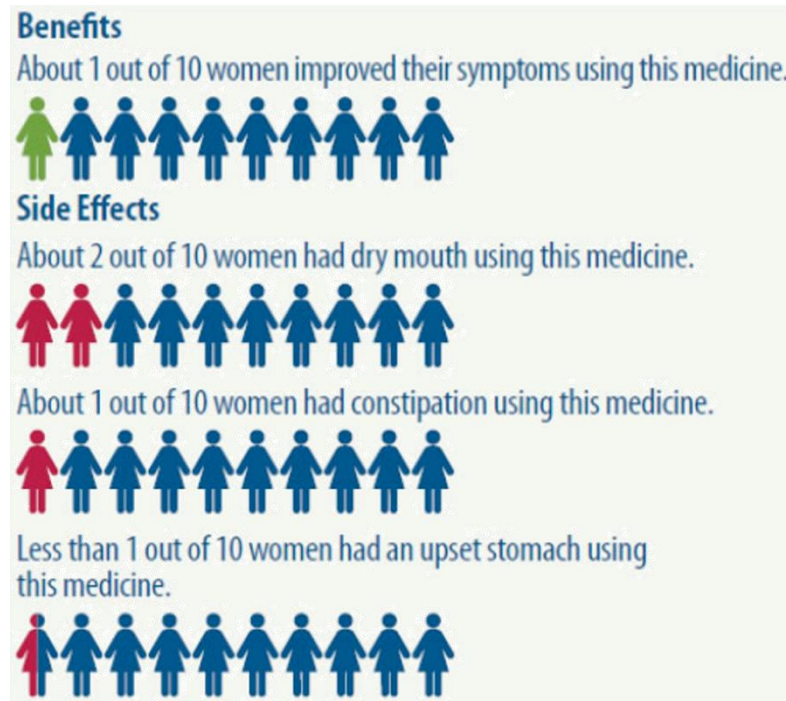
- **Visual Channels**

- **Saturation** of a mark encodes value using *how much color* the mark is drawn with



MARKS & CHANNELS

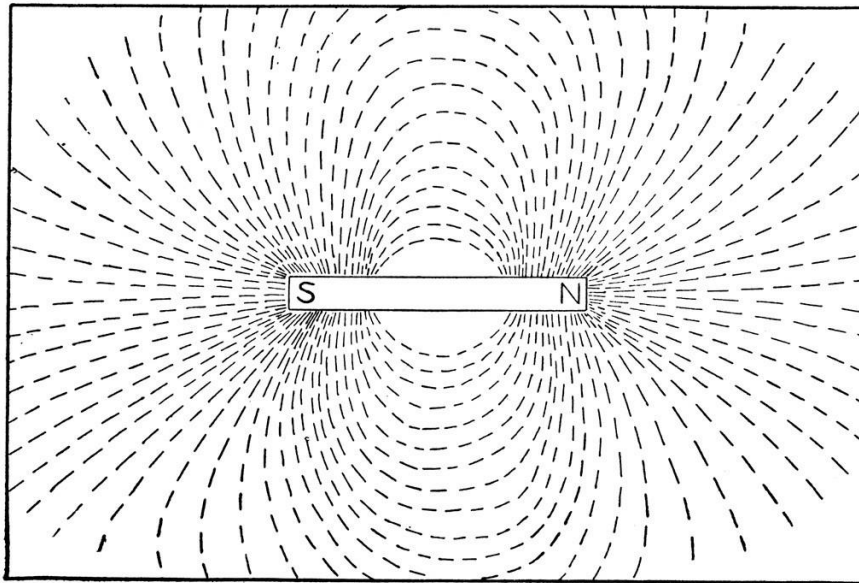
- **Visual Channels**
 - Hue of a mark encodes value using *hue* of the mark



MARKS & CHANNELS

- **Visual Channels**

- Orientation / tile / angle of a mark encodes value using how the mark is *rotated*



MARKS & CHANNELS

- **Visual Channels**

- **Shape** of a mark encodes value using how the mark is *shaped*



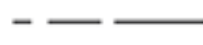
MARKS & CHANNELS

■ Appropriateness of Visual Channels

➡ Magnitude Channels: **Ordered** Attributes

Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance 

Color saturation 

Curvature 

Volume (3D size) 

➡ Identity Channels: **Categorical** Attributes

Spatial region 

Color hue 

Motion 

Shape 

Most

Effectiveness

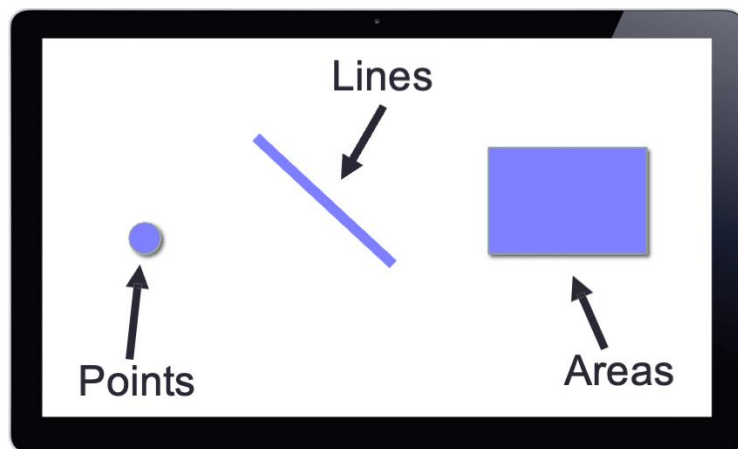
Least

Same

Same

MARKS & CHANNELS

- Find a visualization from your favorite news source
- Identify marks, channels, and what data each channel encodes



➔ Magnitude Channels: Ordered Attributes

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Depth (3D position)	
Color luminance	
Color saturation	
Curvature	
Volume (3D size)	

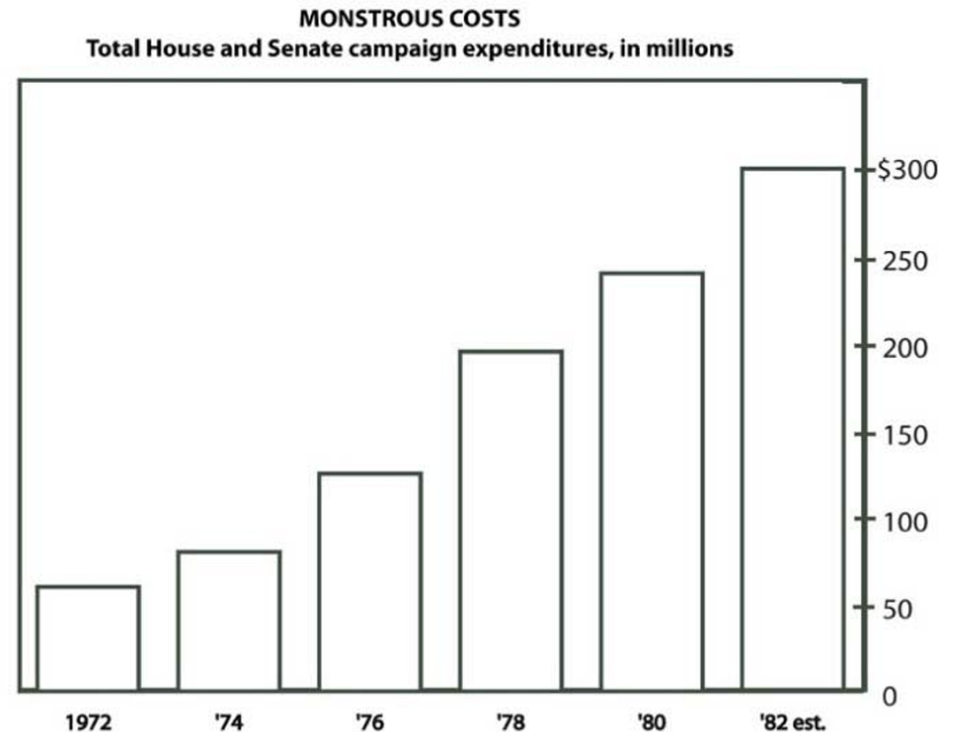
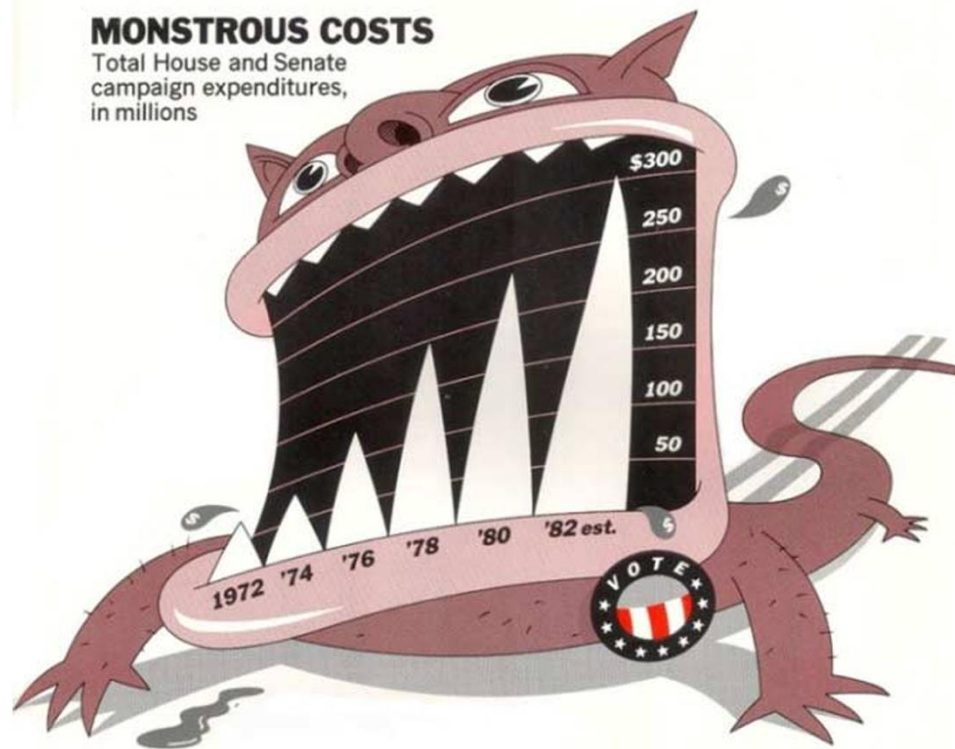
➔ Identity Channels: Categorical Attributes

Spatial region	
Color hue	
Motion	
Shape	

Most
Effectiveness
Least

RULES OF THUMB

- As much as possible, keep it simple (depending on your goal...)

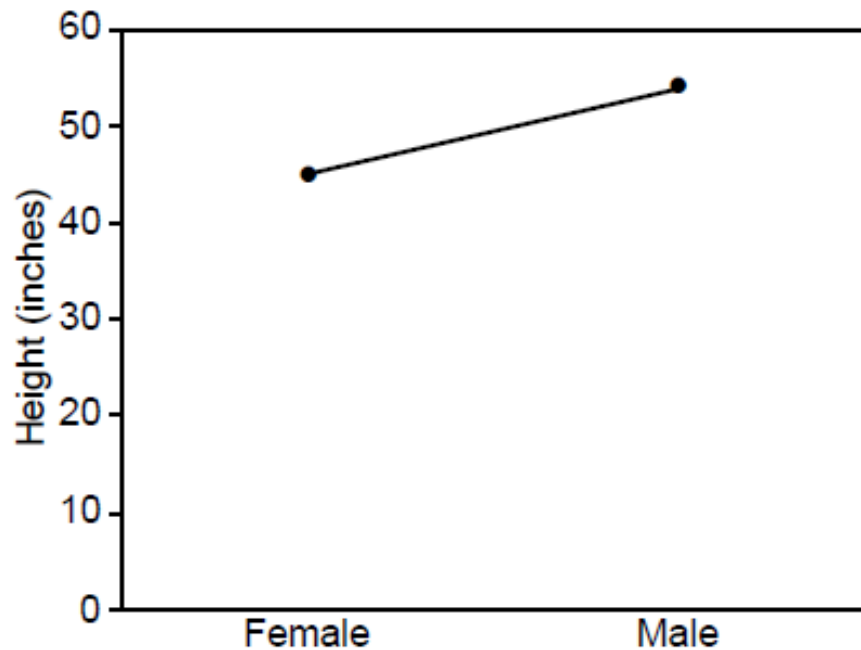


Scott Bateman, Regan L. Mandryk, Carl Gutwin, Aaron Genest, David McDine, and Christopher Brooks. 2010. Useful junk? the effects of visual embellishment on comprehension and memorability of charts.

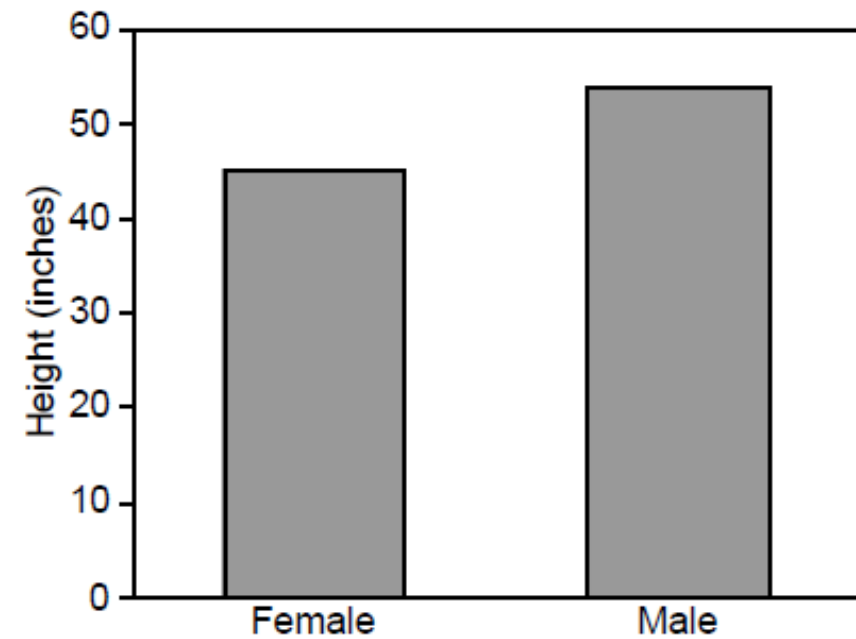
RULES OF THUMB

- Use the right visualization for your data & task

Average Height for Youth Sports Participants

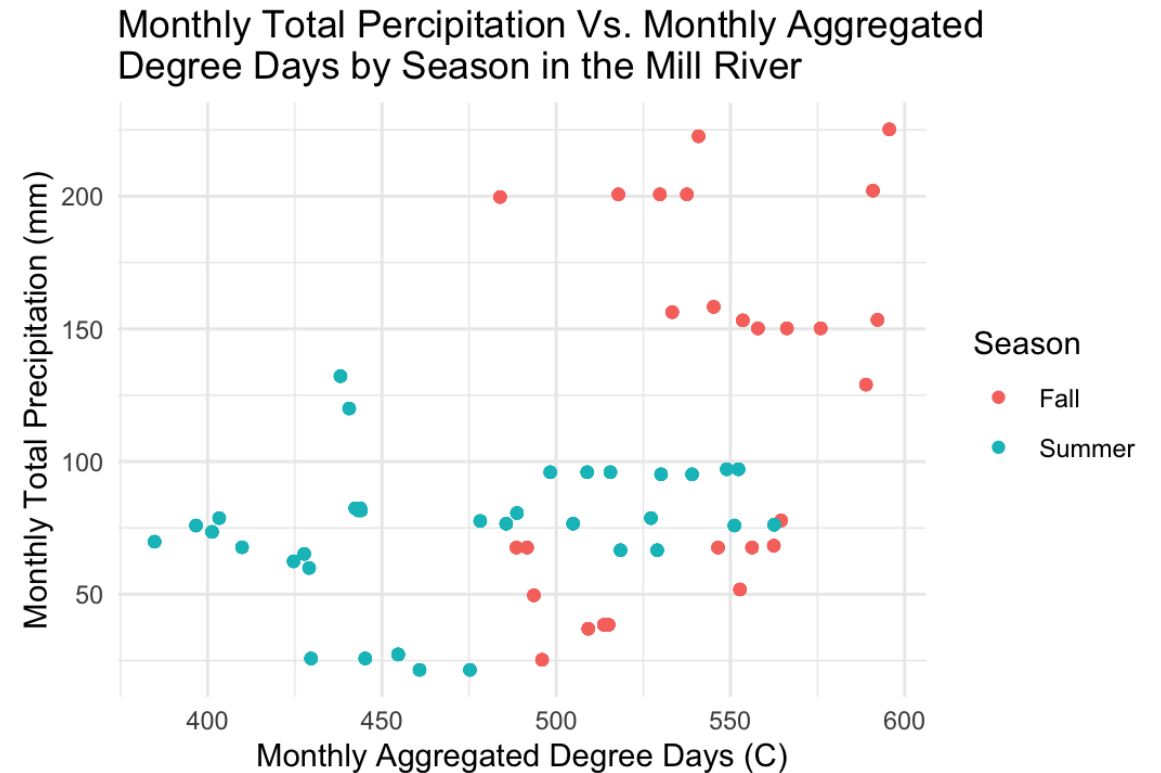
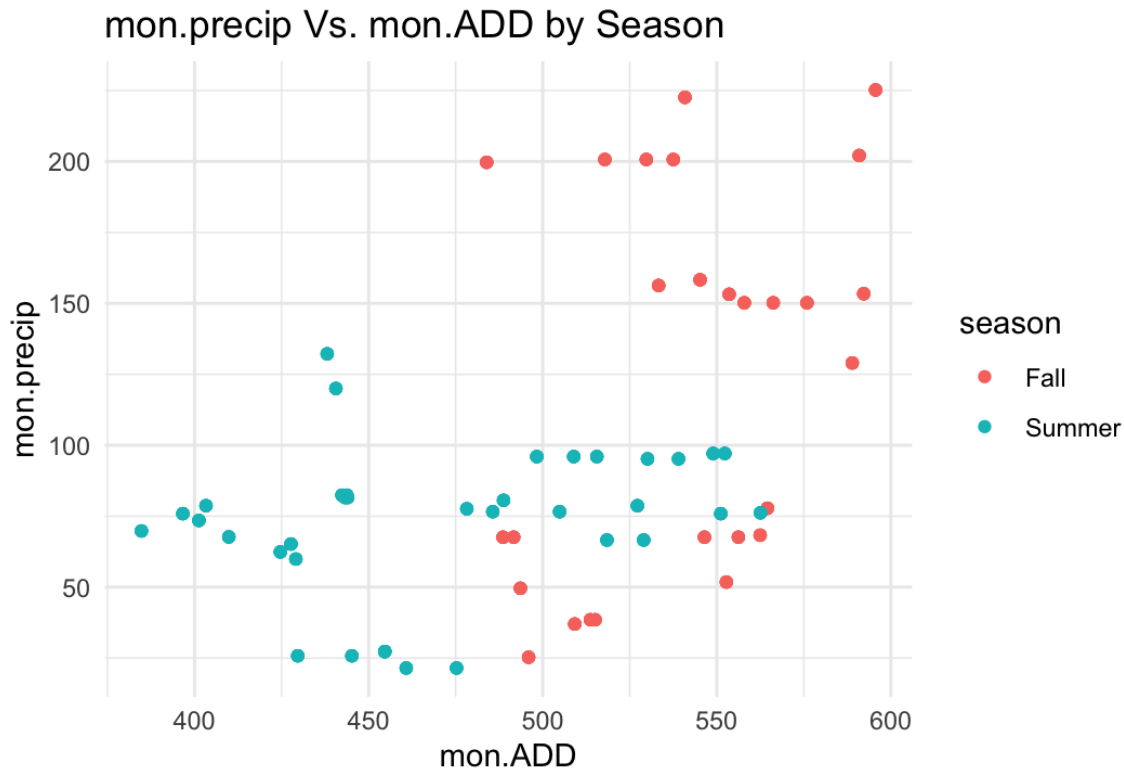


Average Height for Youth Sports Participants



RULES OF THUMB

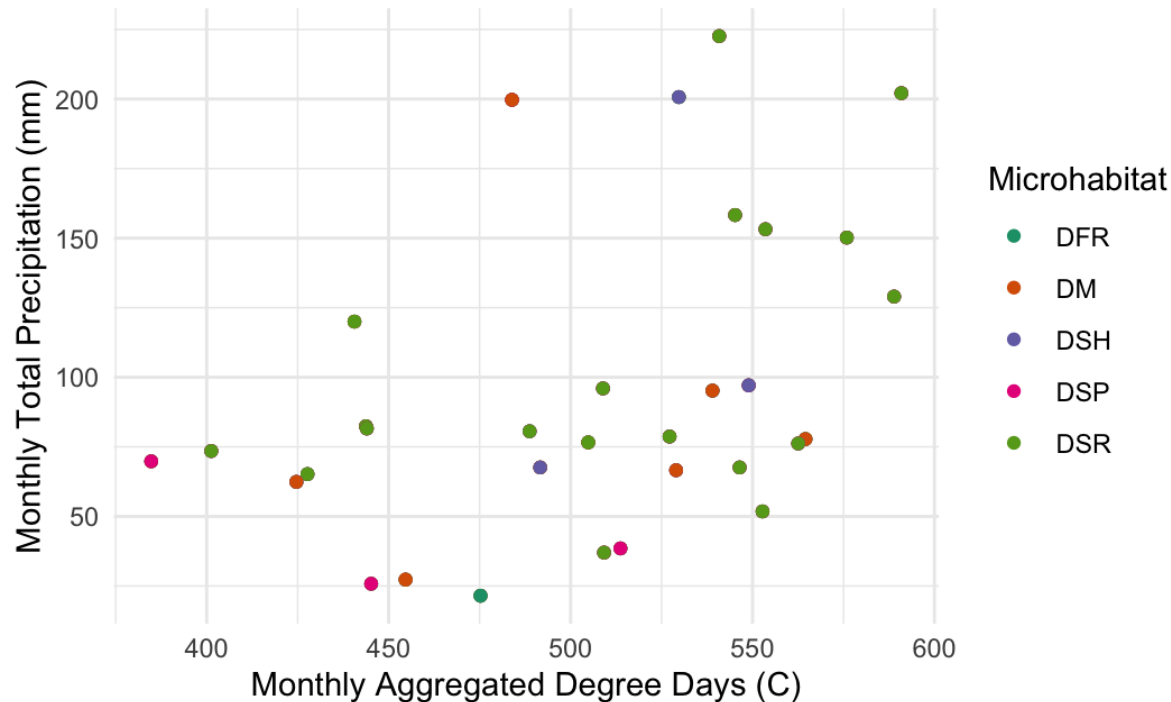
- Clearly title overall visualization, axes, legends, etc.



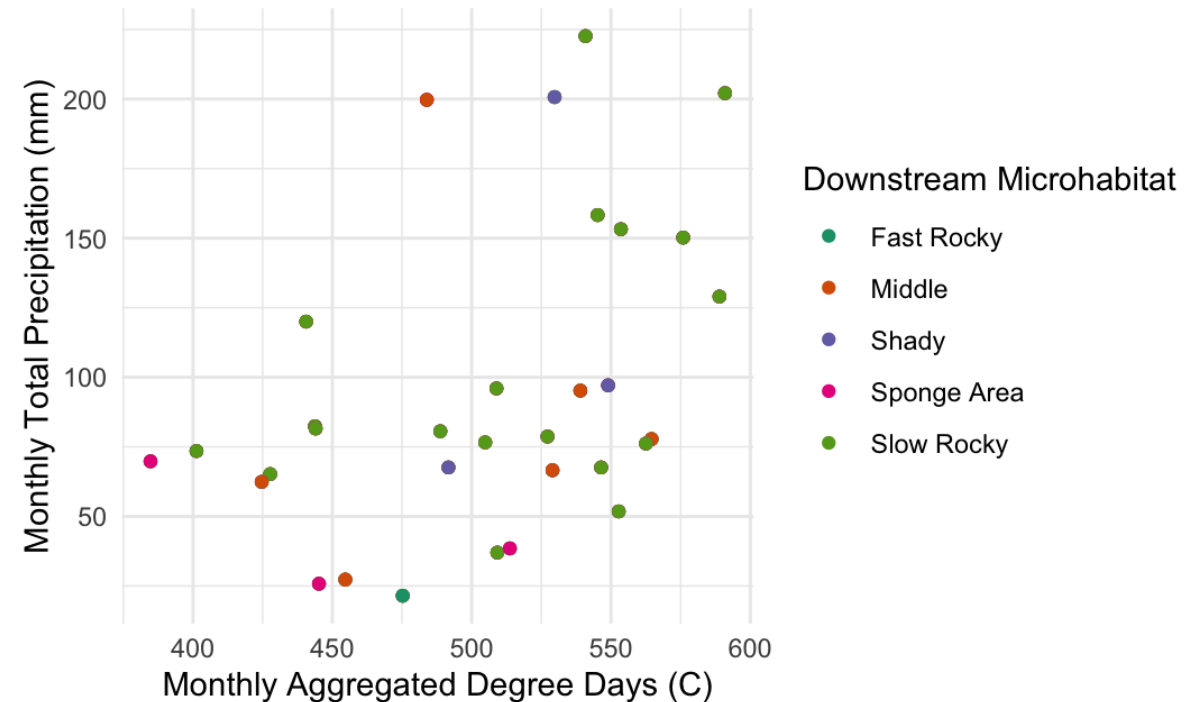
RULES OF THUMB

- Clearly label axes, legends, etc.

Monthly Total Percipitation Vs. Monthly Aggregated Degree Days by Microhabitat in the Mill River

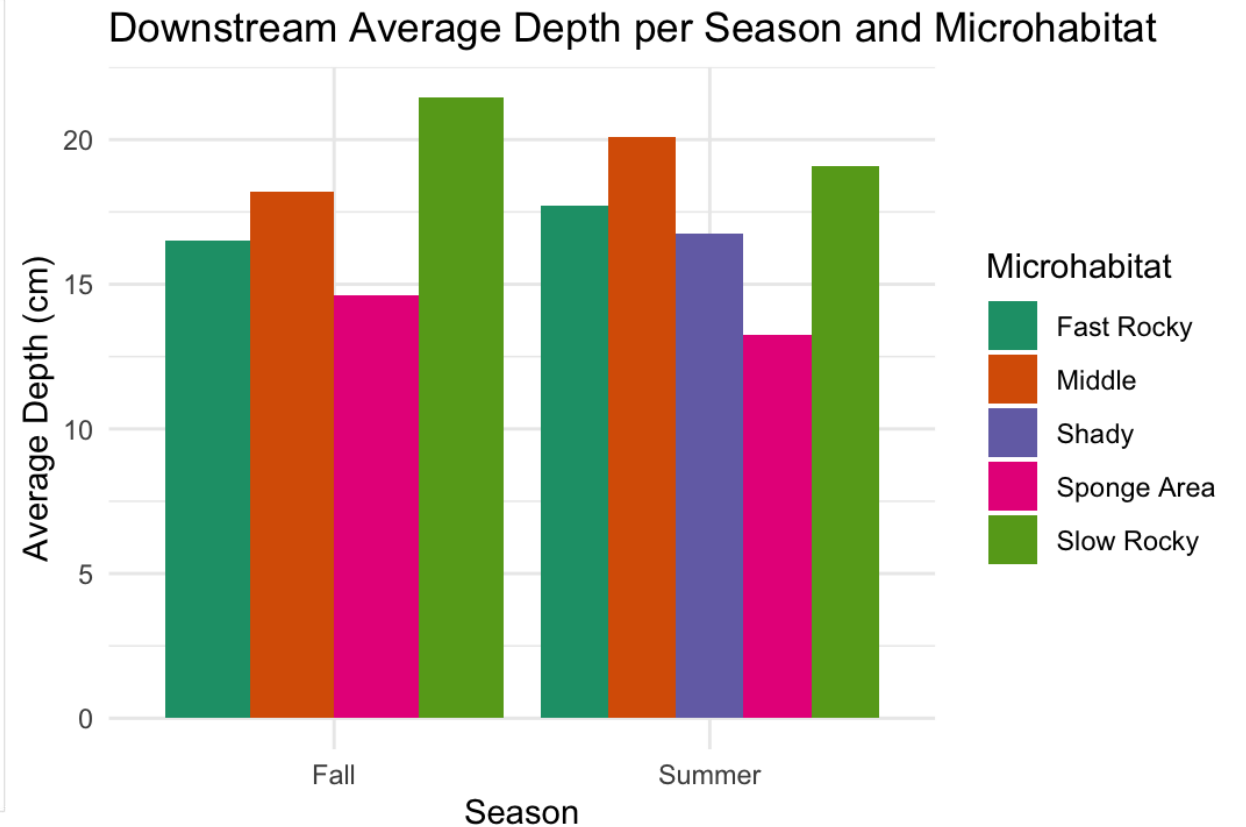
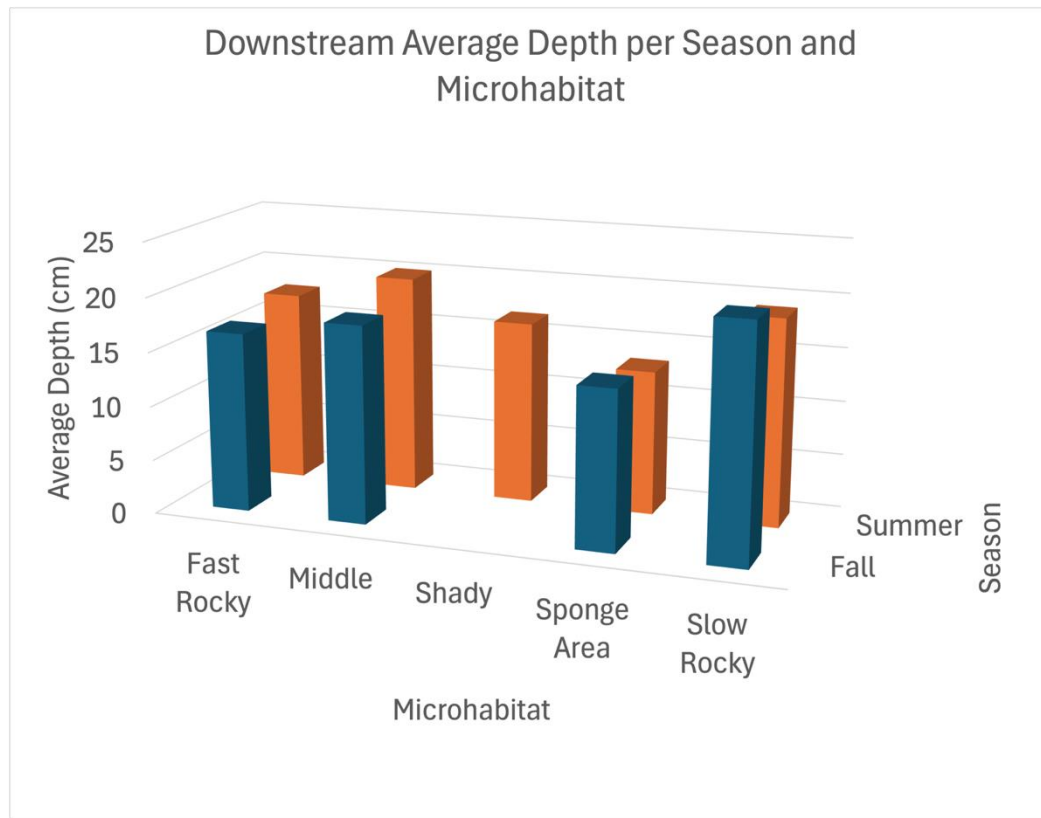


Monthly Total Percipitation Vs. Monthly Aggregated Degree Days by Microhabitat in the Mill River



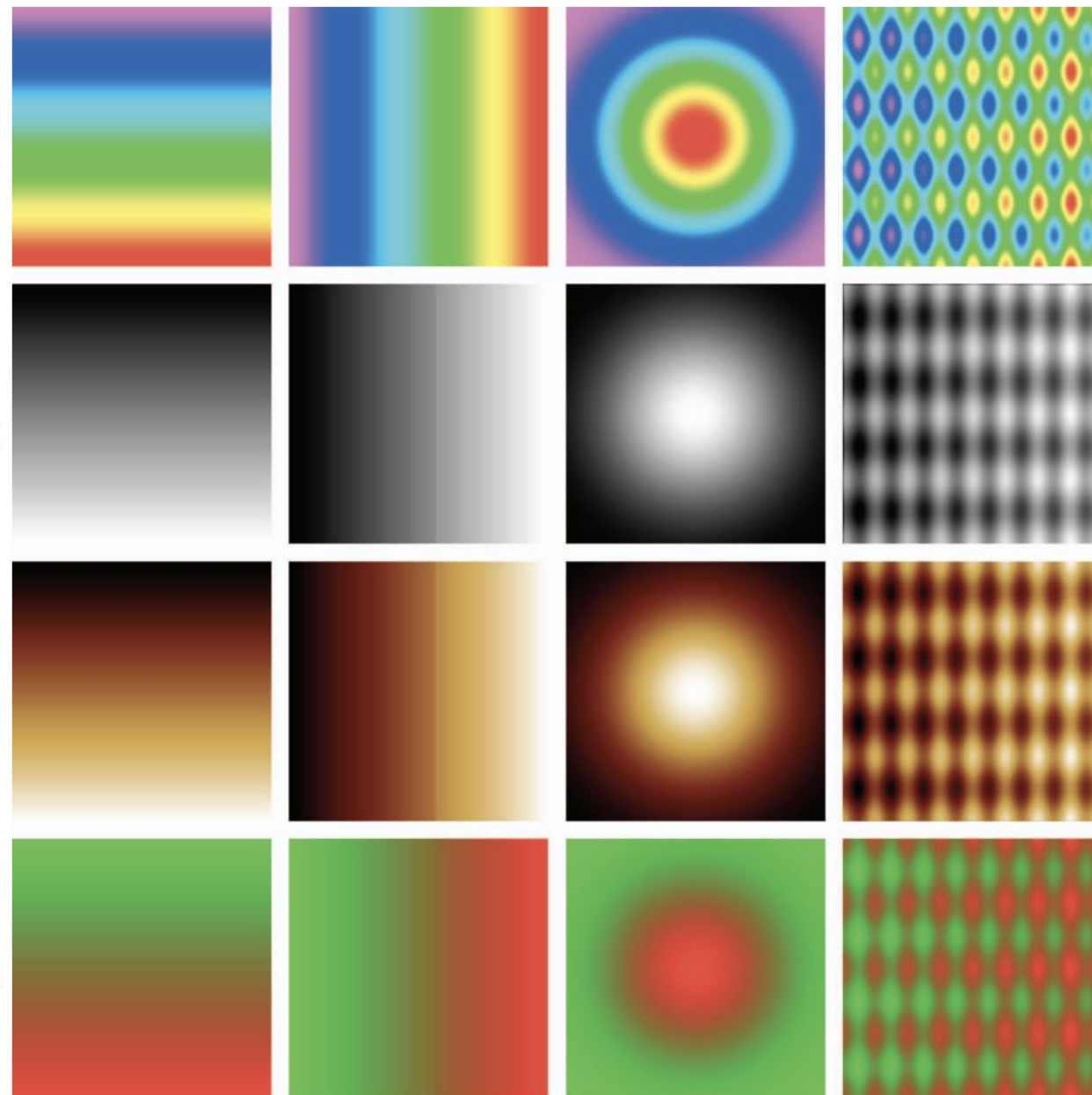
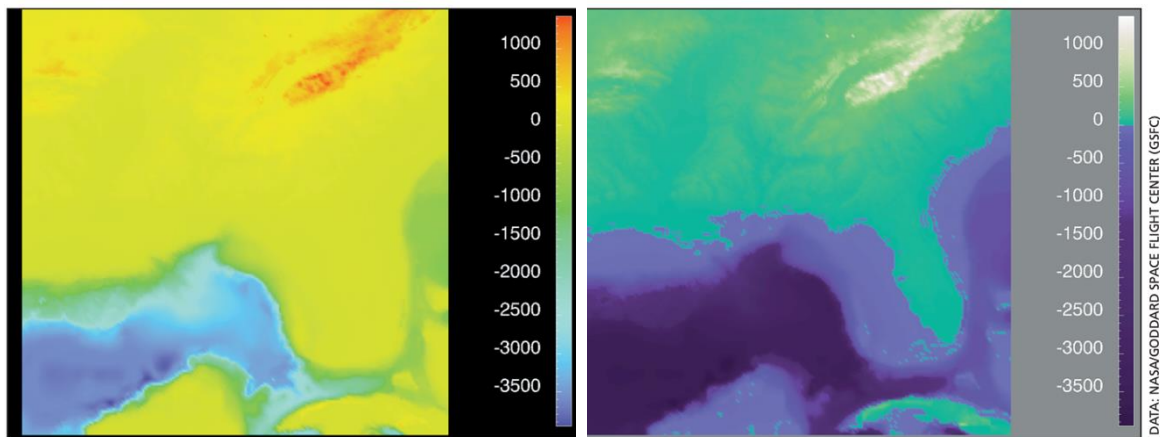
RULES OF THUMB

- Do not use 3D unless necessary



RULES OF THUMB

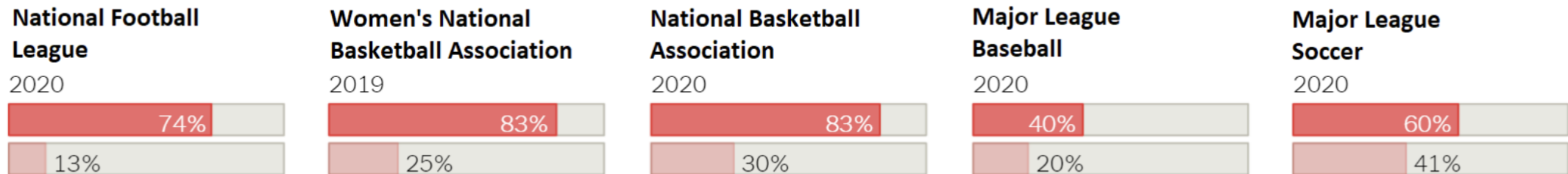
- Use perceptually accurate colormaps



READING VISUALIZATIONS

Racial disparity in major sports leagues

■ Players of color ■ Head coaches of color



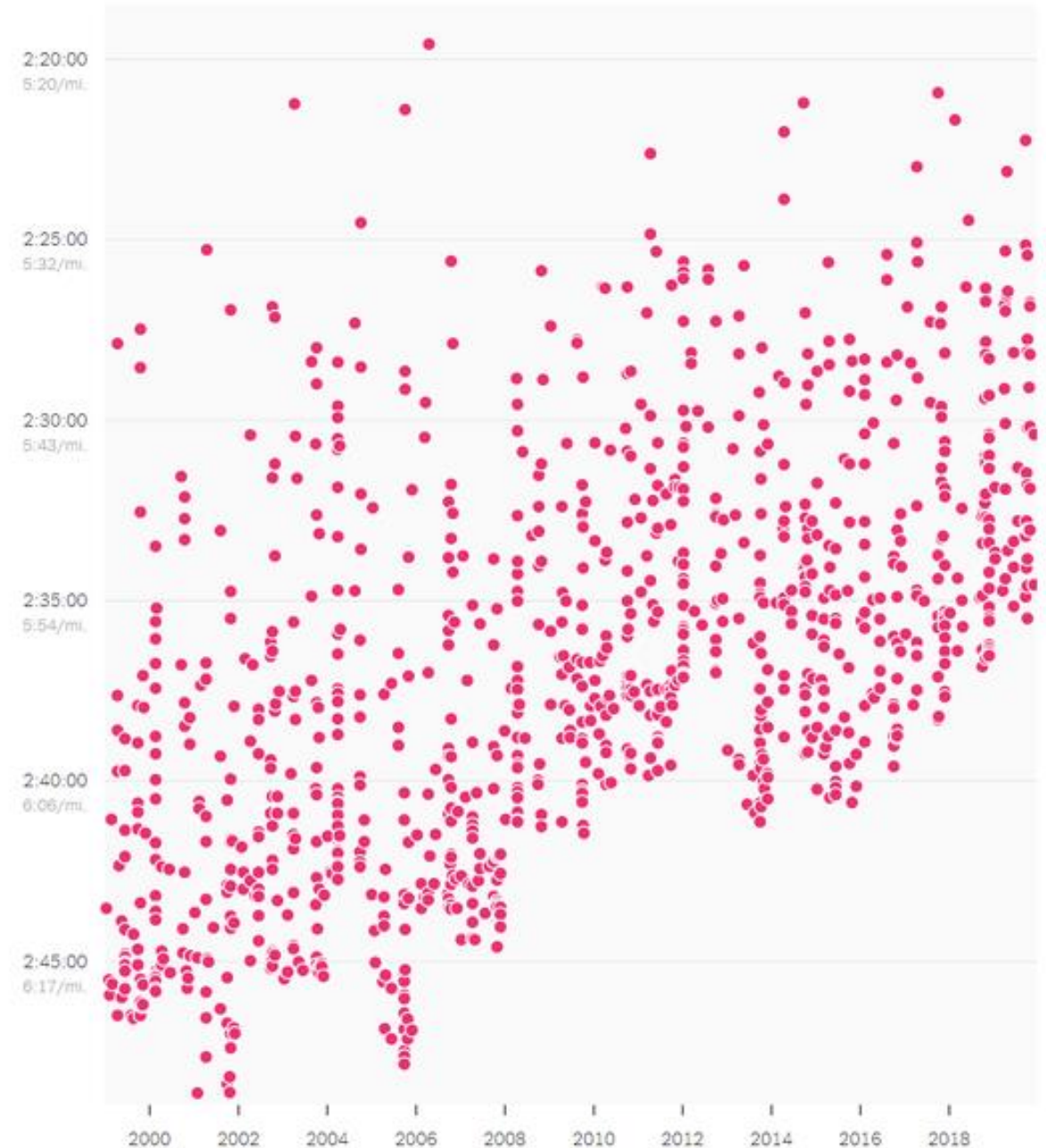
Note: Managers are shown instead of head coaches for M.L.B. The most recent available data is used for each league. The data is self-reported and may not include all league members. ■ Source: Institute for Diversity and Ethics in Sport

<https://www.nytimes.com/2021/01/28/learning/whats-going-on-in-this-graph-diversity-in-professional-sports.html>

READING VISUALIZATIONS

<https://www.nytimes.com/2020/09/17/learning/whats-going-on-in-this-graph-women-marathoners-running-times.html>

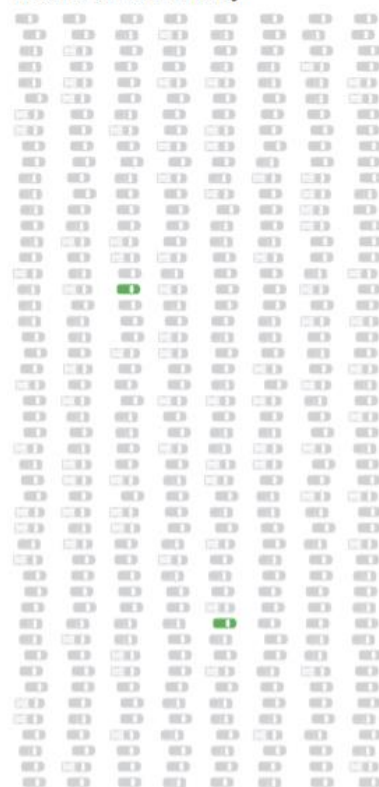
50 fastest U.S. women's marathons per year



READING VISUALIZATIONS

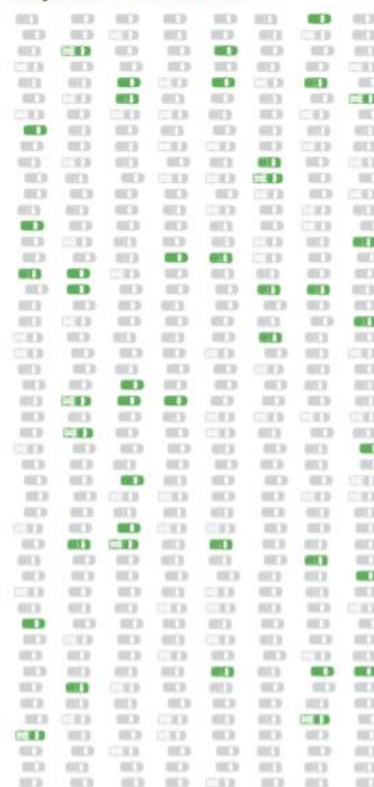
<https://www.nytimes.com/2021/04/15/learning/whats-going-on-in-this-graph-gas-to-electric-vehicle-turnover.html>

Vehicles on the road today



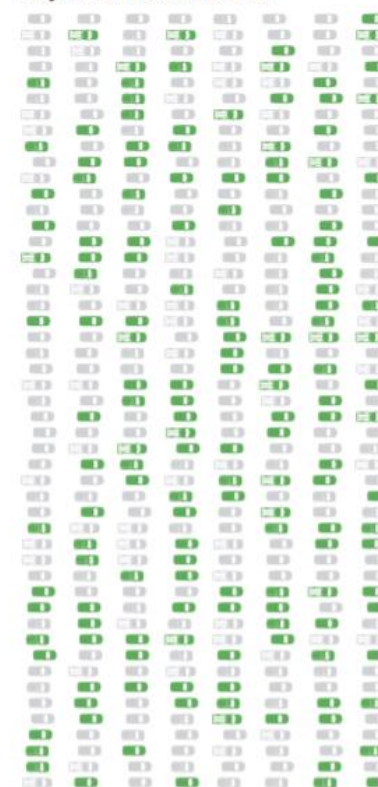
These vehicles represent the 250 million cars, S.U.V.s, vans and pickup trucks on America's roads today. The vast majority run on gasoline. **Fewer than 1 percent are electric.**

Projected on the road in 2035



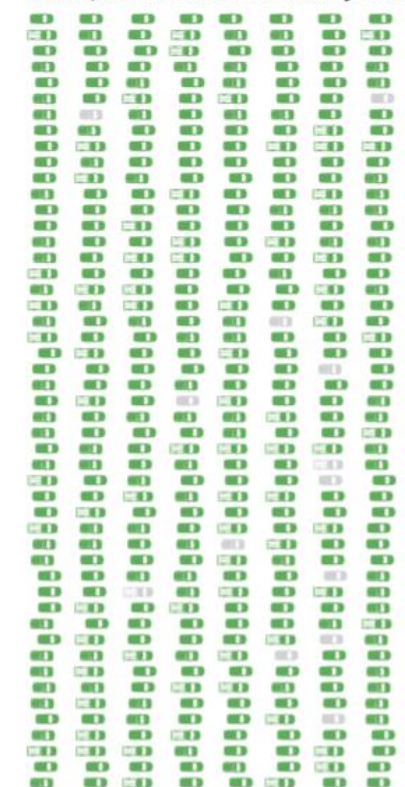
Automakers are now shifting to electric vehicles, which could make up a quarter of new sales by 2035, analysts project. **But at that point, only 13 percent of vehicles on the road would be electric.** Why? Older cars can stick around for a decade or two.

Projected on the road in 2050



Even in 2050, when electric vehicles are projected to make up 60 percent of new sales, **the majority of vehicles on the road would still run on gasoline.** Slow fleet turnover is a major challenge for climate policy.

In 2050, if all new sales are electric by 2035

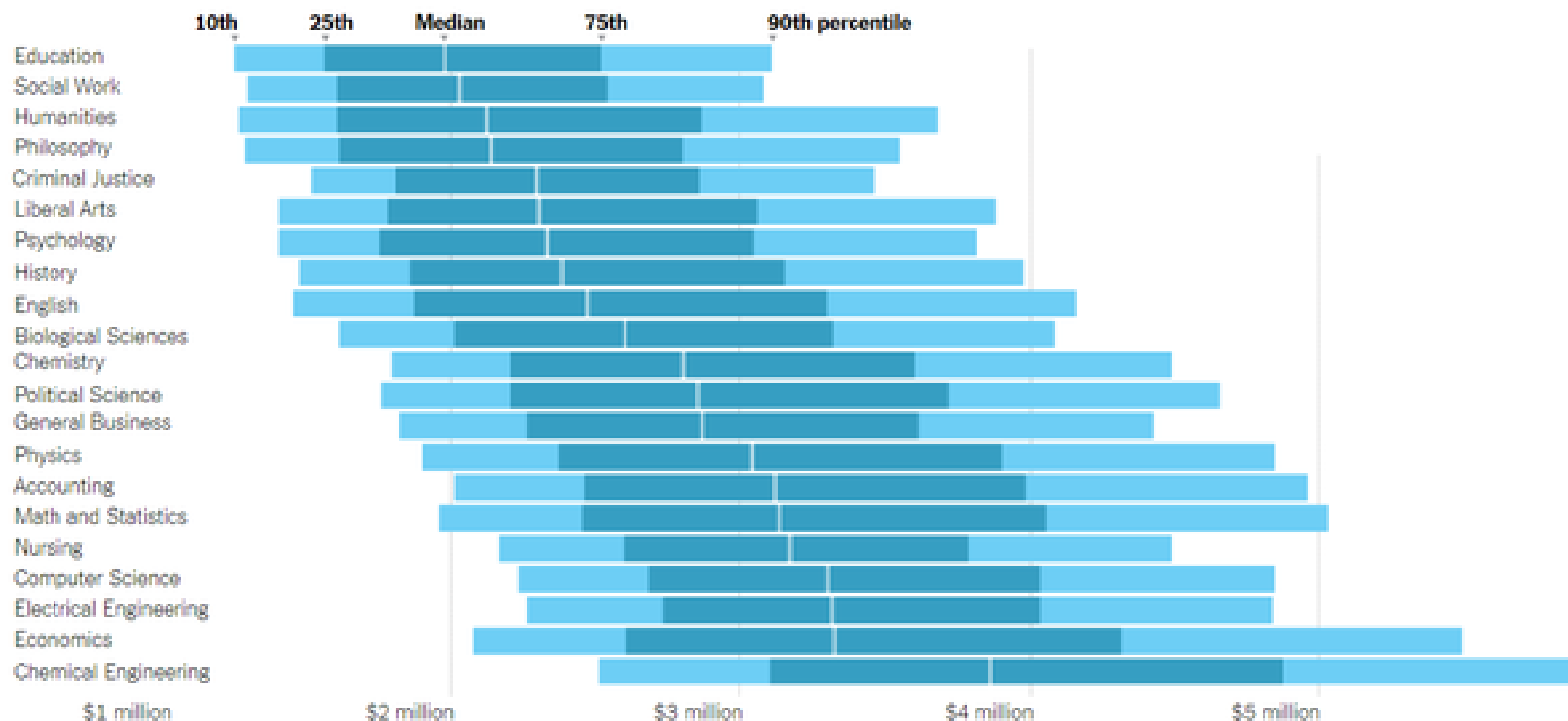


If the United States wanted to **move to a fully electric fleet by 2050** — to meet President Biden's goal of net zero emissions — then sales of gasoline-powered vehicles would likely have to end altogether by around 2035, a heavy lift.

READING VISUALIZATIONS

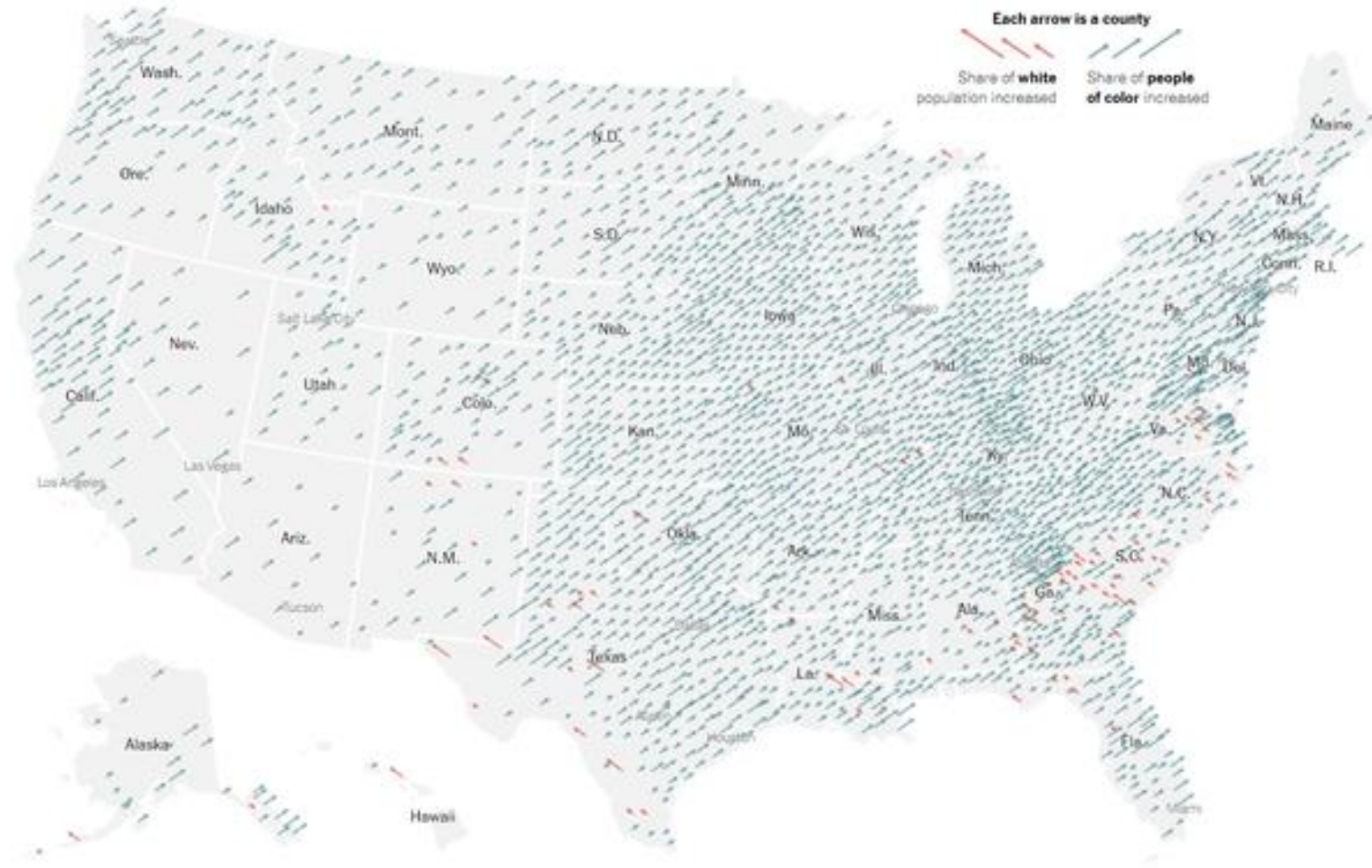
<https://www.nytimes.com/2018/01/04/learning/whats-going-on-in-this-graph-jan-9-2018.html>

Projected career earnings for college graduates in ...



READING VISUALIZATIONS

<https://www.nytimes.com/2022/01/13/learning/whats-going-on-in-this-graph-jan-19-2022.html>



Notes: Arrows on the map show the percentage point change in the share of each group. Counties with a change below one percentage point are not shown. White is defined as non-Hispanic white.

READING VISUALIZATIONS

<https://www.informationisbeautifulawards.com/showcase/7198-the-color-of-sleep>

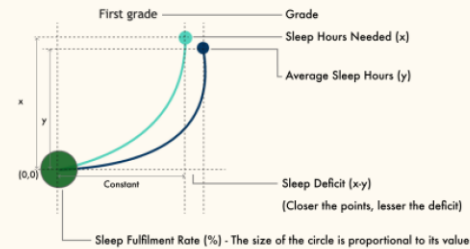
THE COLOR OF SLEEP

Declining Rate of Sleep Fulfilment in students of K-12

As per the data provided by sleepjunkie.com, the visualization studies the gap of sleep hours for students from Kindergarten to 12th grade in terms of Sleep Fulfilment Rate.

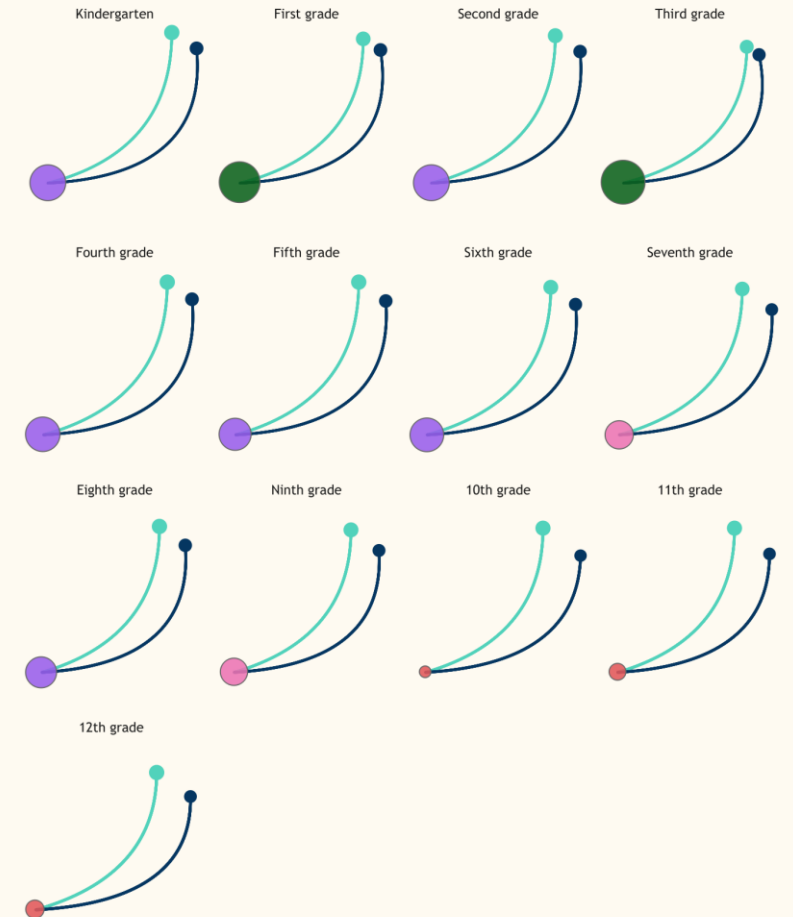
$$\text{Sleep Fulfilment Rate (\%)} = \frac{\text{Average Sleep Hours}}{\text{Sleep Hours Needed}} * 100$$

HOW TO READ THE CHART



Sleep Fulfilment Rate uses a spectrum of four colors to gauge the quality of sleep condition.

Poor Sleep	Moderate Sleep	Better Sleep	Optimum Sleep
< 85	85-87	87-90	> 90
10th grade 81.0%	Ninth grade 85.5%	Eighth grade 87.1%	First grade 92.3%
11th grade 82.1%	Seventh grade 85.9%	Fifth grade 87.6%	Third grade 94.2%
12th grade 82.5%		Sixth grade 88.4%	
		Fourth grade 88.8%	
		Second grade 89.2%	
		Kindergarten 89.5%	

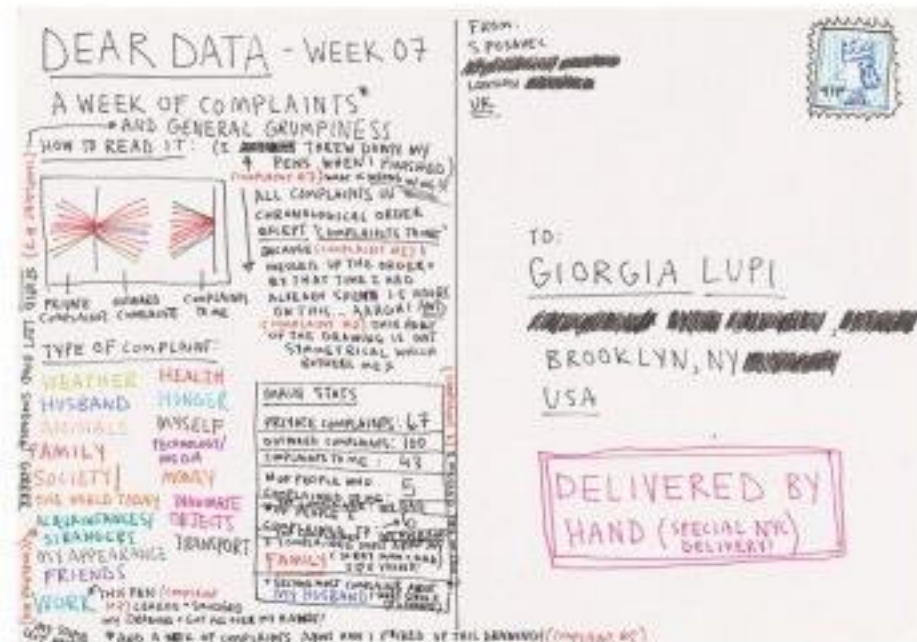
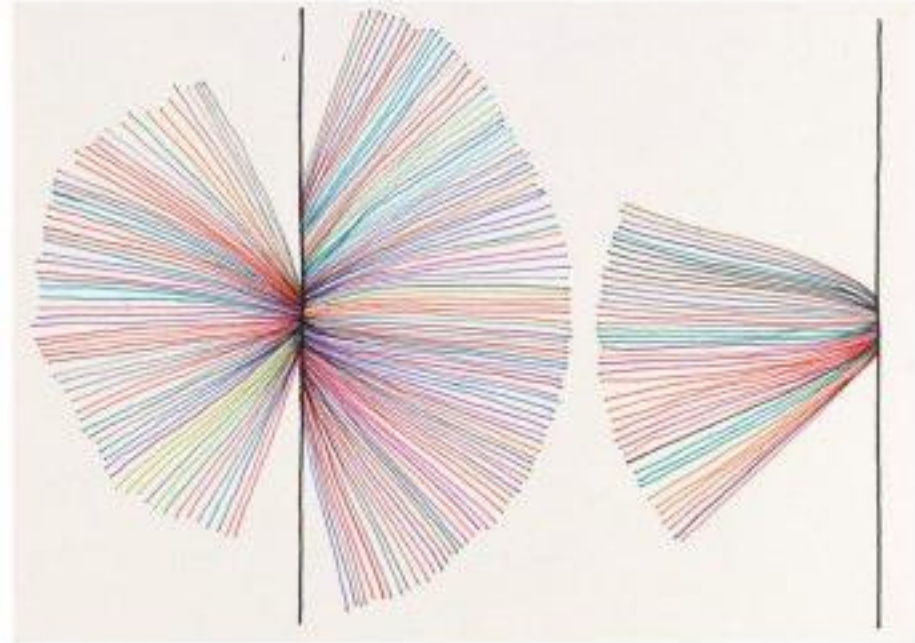


READING VISUALIZATIONS

<https://www.whatdesigncando.com/stories/giorgia-lupi-what-drives-me-is-the-space-between-logic-and-beauty/>

a week of complaints

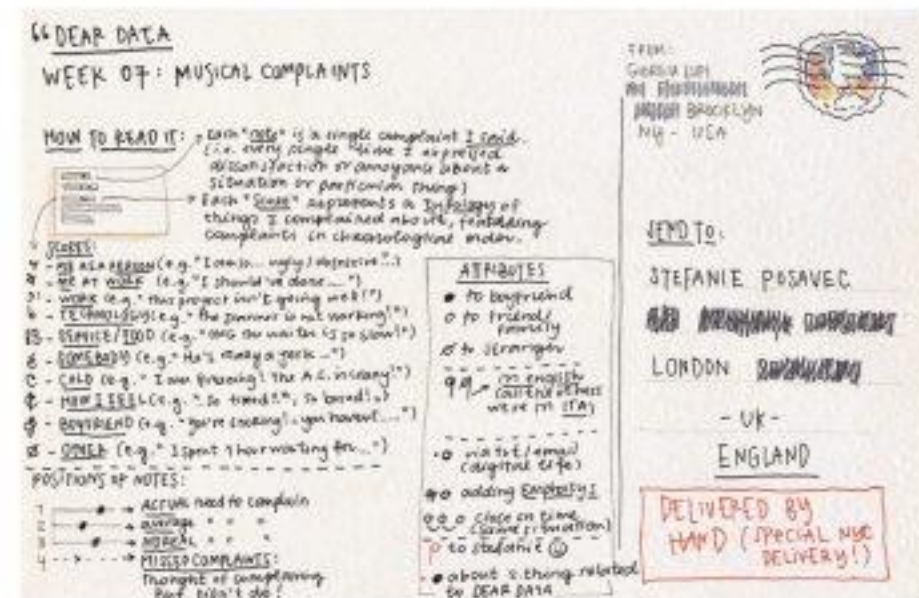
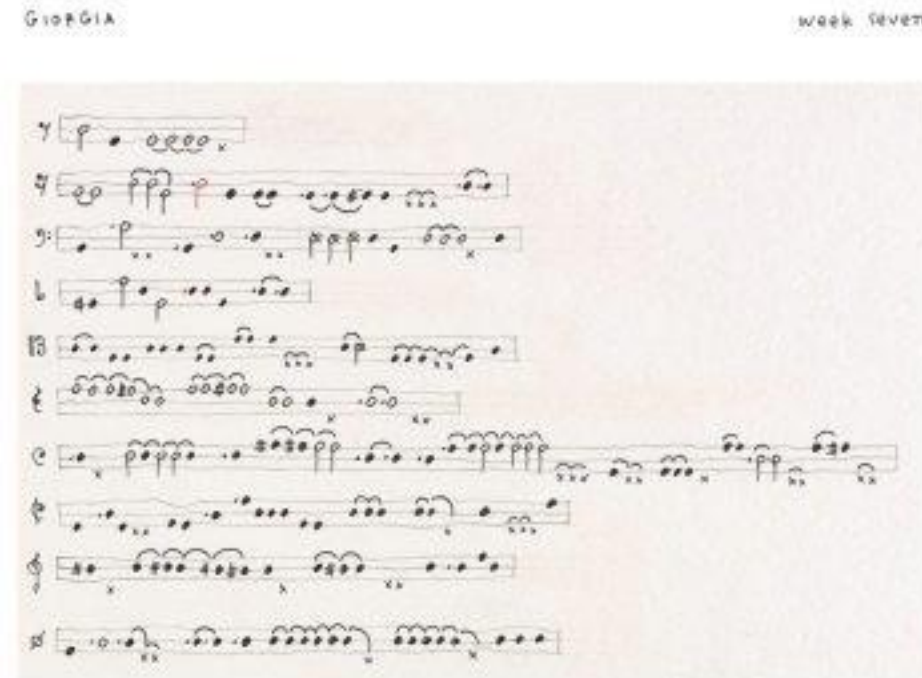
Stefanie



Note the hand-drawn stamps: these postcards were delivered in person in New York!

READING VISUALIZATIONS

<https://www.whatdesigncando.com/stories/giorgia-lupi-what-drives-me-is-the-space-between-logic-and-beauty/>



What better visual reference than a musical score to show the repetitiveness of Giorgio's protests and the "level" of complaint: whether they are justified or totally out of place.

READING VISUALIZATIONS

Explore one of the following:

- <https://vega.github.io/voyager2/>
- <https://lumos-webapp-4aeadb3bf30d.herokuapp.com/>
- <https://public.tableau.com/newWorkbook/3e5b7248-10c4-48c6-9abe-83eeda8908d3#1> (you will need a free account and to upload a .csv)

How do these tools incorporate visualization?